

Isadora Santos

AI & Backend Engineer

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PROFESSIONAL SUMMARY

AI & Backend Engineer with 5+ years of enterprise experience specializing in agentic AI workflows, backend modernization, and secure pipeline automation. Proven expertise in building reliable LLM applications, adaptive memory systems, and scalable infrastructure to automate complex operational tasks with rigorous software engineering practices.

TECHNICAL SKILLS

- **AI & Machine Learning:** Python, LangChain, LangGraph, LangSmith, ChromaDB, PostgreSQL pgvector, OpenAI API, Gemini API, Claude.
- **Engineering & MLOps:** Docker, Kubernetes (AKS/EKS), MLflow, Apache Airflow, FastAPI.
- **Backend & Databases:** Python, Java, SQL, JavaScript.
- **Cloud & Security:** AWS, Azure, SecOps, HashiCorp Vault, REST APIs, Microservices.

EDUCATION

- **MLOps - Machine Learning Operations** | Duke University | Present
- **Artificial Intelligence Specialist Diploma** | GRAN | Present
- **Google AI Professional Certificate** | In Progress
- **Systems Analysis and Development Diploma** | UNINASSAU
- **Computer Programming Diploma** | Georgian College
- **Specialization in IT Management** | Gran Concursos

FEATURED AI APPLICATIONS & SecOps PROJECTS

Stateful AI Vulnerability Agent (Adaptive Memory System)

- Architected an open-source agentic workflow using LangGraph and the Google Gemini API to automate complex triage and decision-making tasks, inspired by enterprise vulnerability management systems.
- Engineered a containerized FastAPI and Docker backend featuring a ChromaDB Retrieval-Augmented Generation (RAG) system, utilizing PostgreSQL pgvector for reliable state memory across expert feedback loops.

Secure MLOps Pipeline

- Developed a secure CI/CD MLOps pipeline to manage anomaly detection models, integrating continuous code quality checks (flake8) and automated security scanning (Trivy) via GitHub Actions.
- Established consistent model tracking and pipeline execution by containerizing the training environment and utilizing enterprise deployment standards like MLflow and LangSmith.

Agentic Data Extraction & Self-Healing ETL

- Built an intelligent, LangGraph-driven ETL pipeline capable of ingesting unstructured, messy data inputs and dynamically mapping them into strict analytical schemas.
- Developed an interactive React frontend designed for human-in-the-loop validation, allowing non-technical specialists to seamlessly review and correct the agent's extraction logic, backed by a PostgreSQL data persistence layer.

EXPERIENCE

Independent AI & DevSecOps Engineer | Montreal, QC | May 2025 - Present

- Architected production-grade AI applications utilizing Python, LangGraph, and LLMs, backed by secure PostgreSQL infrastructure and enterprise CI/CD practices (Trivy, MLflow, LangSmith).
- Developed *DDS na palma da mão*, an interactive WebAR workplace safety application using A-Frame and MindAR, translating specialized safety protocols into intuitive digital formats for end-users.
- Built seamless integrations connecting intelligent agents to persistent databases and external APIs to simulate complex enterprise environments.

RBC - ROYAL BANK OF CANADA (Capital Markets) | Senior Engineer | Montreal, QC | Apr 2024 - Apr 2025

- Architected a Java Spring Boot and Azure OpenAI "AI Gateway" serving 10+ distinct teams, acting as the integration layer between core infrastructure and LLM capabilities.

- Led the legacy migration of monolithic settlement systems to Azure Kubernetes Service (AKS) microservices, managing complex integrations with highly specialized financial backend services.
- Designed a secure Jenkins and MLflow continuous training pipeline for risk models enforcing strict Role-Based Access Control (RBAC), establishing automated feedback loops and reproducible deployment standards.

RBC - ROYAL BANK OF CANADA (Technology & Operations) | SecOps Engineer / DevSecOps | Toronto, ON
| Jan 2022 - Apr 2024

- Architected and built a vulnerability management RAG assistant, automating threat triage and ticket generation architecture.
- Built an Apache Airflow, SQL, and AWS Lambda orchestration layer that automated enterprise vulnerability management, cutting Mean Time to Remediate (MTTR) by 60%.
- Developed 'Self-Healing' Python infrastructure bots and an internal Cortex XSOAR cloud notification system, automatically correcting misconfigured cloud resources and reducing incident response times by 60%.

MUNICIPALITY OF YORK REGION | Statistical Data Analyst | Newmarket, ON | Jan 2021 - May 2021

- Developed automated Python data pipelines and secure ETL processes for sensitive health data (PHI), ensuring strict compliance with privacy regulations.
- Engineered robust data extraction workflows to dynamically map complex, unstructured public health information into reliable schemas for reporting and analysis.
- Engineered a COVID-19 screening automation application using Microsoft PowerApps, SharePoint, and Power BI, managing the agile lifecycle and reducing manual errors by 70%.

LANGUAGES

- English: Fluent
- Portuguese: Fluent/Native
- Spanish: Advanced
- Russian: Intermediate
- French: Basic (currently learning)